

Exam 3 — Own Machine Learning Project

Theoretical Question 1

Compare **linear models**, **CNNs**, and **RNN/Transformers** regarding:

1. typical data types
2. strengths and weaknesses
3. overfitting risks and regularization strategies

Provide one concrete use case for each model family.

Theoretical Question 2

Explain how neural networks are trained with **gradient descent**:

1. role of the loss function and its gradient
2. batch vs. mini-batch vs. stochastic gradient descent
3. what the learning rate controls, and what happens if it is too large or too small
4. what is the difference between gradient descent and gradient ascent, and in which situations is each one used?

Project Task — Student’s Own ML Project

Design and implement your **own ML project** (topic and dataset chosen by you).

Mandatory requirements

1. **Problem relevance**: Explain why the problem is relevant (technical/business/social context).
2. **Data choice**: Explain why you selected this dataset, including source, size, quality, and expected limitations.
3. **Model setup**: Justify your architecture and preprocessing pipeline.
4. **Baselines**: Compare against at least one simpler baseline.
5. **Evaluation**: Use suitable metrics and discuss failure cases.
6. **Interpretation**: Explain what your model learned and where it can fail.

Deliverables

- 5-8 page report
- reproducible code/notebook
- short conclusion: “Why this data and setup were the right choice (or not)”